Feature Relationship Instability

Derived feature Instability

Two different "instabilities" concepts

1. **Explanatory - Dependent instability (predictive relationship instability)**
   - The relationship between the explanatory and the dependent variable change over time and/or across regimes

2. **Explanatory - Explanatory instability (internal structure instability)**
   - Correlation/collinearity between explanatory variables drifts
     - **Collinearity drift:** two features become "the same thing" during crisis (2008), but are independent in normal times.
   - Lead–lag relationships among explanatory variables shift - who leads whom changes
   - The latent factor structure of the explanatory set changes (feature loadings rotate)

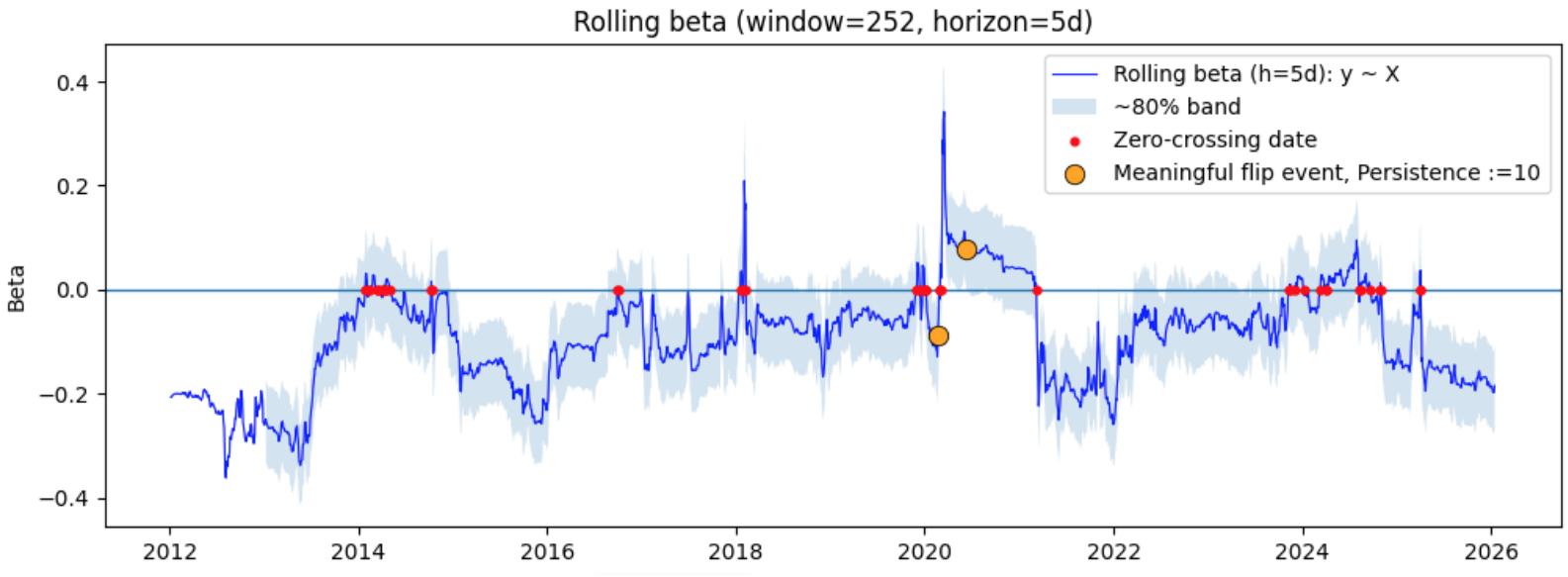
In multivariate modelling

- Performance depends **not only on how each explanatory variable relates to the dependent variable**, but also on the **explanatory variables' joint behaviour P(X)**

Subtle consideration on joint structures $\mathbf{P}(\chi)$

- **Nonlinear dependence:** features can be related in a curved/threshold way even when linear correlation is low
- **Tail dependence:** features may co-move mainly not in normal times
- **State-conditional dependence:** relationships hold only in certain regimes (high-vol, low-vol)
- **Distribution shift:** individual feature distributions change

Simplified example – Rolling Beta



$$\hat{\rho}_t^{(W)} = \mathrm{Corr}\big(\{X_{1,\tau}\}_{\tau=t-W+1}^{t},\ \{X_{2,\tau}\}_{\tau=t-W+1}^{t}\big)$$

Rolling β_t shows the **time-varying explanatory effect** of the past 5-day return on the next 5-day return; sign and magnitude changes indicate **relationship instability**

$$\mathbb{E}[\,y_t \mid X_t\,] \approx a_t + \beta_t X_t$$

$$X_t = \frac{P_t}{P_{t-5}} - 1 \quad \text{(last 5 days)}$$

$$y_t = \frac{P_{t+5}}{P_t} - 1 \quad \text{(next 5 days)}$$

Instability Considerations

The implications if instability not identified and addressed

- Confidence from backtests (performance driven by one historical regime)
- Feature importance might be non-transferable from train to test
- Model not robust – significant exogenous changes
- Spurious features (proxy effects disappear when underlying mechanism changes)

Instability Impacts are Model Agnostic

Instability is a property of the data-generating process, not the estimator

- Under these conditions models trained on one regime can mis-specify another
  - $\mathbf{P}$ (y| χ) - the mapping from explanatory to dependent
  - $\mathbf{P}(\chi)$ - the joint behaviour of the explanatory variables

Potential remediation methods

- Walk-forward training
- Regime-aware models
- Time-varying parameter models
- Explicit drift detection and retraining trigger
- Other…